

CYCLE-AWARE FORMAN-RICCI CURVATURE FOR SIGNED & BIPARTITE NETWORKS*

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Abstract. This document outlines the mathematical framework and empirical validation for the Cycle-Aware Forman-Ricci Curvature (F^*) applied to signed and bipartite networks. We resolve the metric collapse of standard Forman-Ricci Curvature in networks lacking 3-cliques by incorporating higher-order cycles (C_4), structural balance weighting, and a strictly bounded $\mathcal{O}(d)$ degree normalization to prevent combinatorial explosion in dense datasets.

Key words. Cycle-Aware Curvature, Forman-Ricci Curvature, Signed Networks, Bipartite Graphs, Algebraic Topology

AMS subject classifications. 05C22, 05C38, 05C50, 05C82

1. Introduction. Forman-Ricci Curvature (FRC) is a discrete geometric invariant that evaluates the structural cohesiveness of networks by measuring the extent to which edges are embedded within higher-order cliques, primarily triangles. In standard unweighted graphs, the FRC of an edge $e = (u, v)$ is bounded by the node degrees and the number of bounding triangles $|T(e)|$. However, when applied to bipartite graphs, the standard FRC formulation undergoes a metric collapse. Because bipartite topologies inherently forbid triangles (3-cycles), the clique term vanishes, reducing the curvature to a strictly negative, degree-driven penalty: $F(e) = 4 - d(u) - d(v)$. Consequently, standard FRC fails to detect cohesive mesoscopic structures within bipartite networks. To resolve this, we introduce a Cycle-Aware formulation that leverages 4-cycles (C_4) while enforcing strict $\mathcal{O}(d)$ normalization to prevent combinatorial run-time explosion in dense real-world datasets.

2. Mathematical Formulation.

DEFINITION 2.1 (Cycle-Aware Forman-Ricci Curvature for Signed Bipartite Networks). *Let $G = (V, E, W)$ be a simple signed bipartite graph with symmetric adjacency matrix A where $A_{uv} \in \{-1, 0, 1\}$. For an edge $e = (u, v) \in E$, let $d(u)$ and $d(v)$ denote the unweighted degrees. Let $C_4(e)$ denote the set of 4-cycles containing e . The structurally balanced weight of a cycle $C \in C_4(e)$ is determined by $\text{sgn}(C) = \prod_{e' \in C} A_{e'}$. To prevent quadratic $\mathcal{O}(d^2)$ unboundedness in dense networks, the cumulative cycle term is scaled by the theoretical maximum cycle bound $\max(1, (d(u) - 1)(d(v) - 1))$. The Cycle-Aware Forman-Ricci Curvature $F^*(e)$ is defined as:*

$$(2.1) \quad F^*(e) = 4 - d(u) - d(v) + \gamma \min(d(u) - 1, d(v) - 1) \left[\frac{\Sigma_{C_4}(u, v)}{\max(1, (d(u) - 1)(d(v) - 1))} \right]$$

where $\gamma = 3$ is a scaling constant mirroring standard FRC properties.

LEMMA 2.2 (Non-Overcounting of 4-Cycles). *Let $e = (u, v) \in E$. The net 4-cycle weight summation $\Sigma_{C_4}(u, v) = \sum_{C \in C_4(e)} \prod_{e' \in C} A_{e'}$ evaluates each 4-cycle exactly once without multiplicity via the sparse matrix relation:*

$$(2.2) \quad \Sigma_{C_4}(u, v) = A_{uv}(A^3)_{uv} - d(u) - d(v) + 1$$

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Proof. The matrix entry $(A^3)_{uv} = \sum_{x,y \in V} A_{ux}A_{xy}A_{yv}$ enumerates the sum of edge weight products for all walks of length 3 between u and v . In a simple bipartite graph, degenerate walks take three explicit forms: 1) $u \rightarrow v \rightarrow x \rightarrow v$, 2) $u \rightarrow y \rightarrow u \rightarrow v$, and 3) $u \rightarrow v \rightarrow u \rightarrow v$. Summing these contributions yields $(d(u) + d(v) - 1)A_{uv}$. Any non-degenerate simple path (u, y, x, v) corresponds to exactly one unique 4-cycle $C \in \mathcal{C}_4(e)$. By isolating the simple paths and multiplying the expansion of $(A^3)_{uv}$ by A_{uv} , we obtain $A_{uv}(A^3)_{uv} = \Sigma_{C_4}(u, v) + d(u) + d(v) - 1$. Rearranging this algebraic identity yields (2.2), extracting the exact net signed contribution of all 4-cycles without duplication. $\square$

3. Empirical Validation.

3.1. Heterogeneous Block Model. The algorithm was validated on a non-homogeneous synthetic block model possessing diverging subset densities. The calculated F^* values confirm algebraic validity, successfully separating dense homogeneous sub-structures from sparse, negatively-weighted topological bridges, as shown in Figure 3.1.

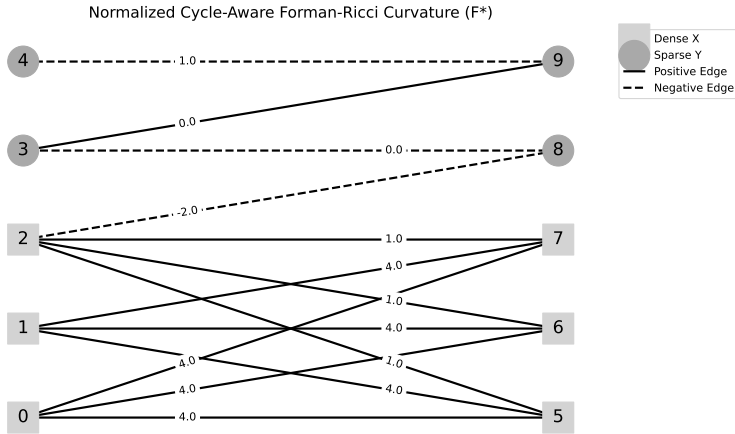


FIG. 3.1. *Topology of the synthetic non-homogeneous signed graph. Nodes are assigned shapes by cohesive groups (squares = dense subset, circles = sparse subset). Edges are styled by their sign (solid = positive, dashed = negative). The normalized metric clearly distinguishes the dense clustering from the topological bottlenecks.*

3.2. Large-Scale Real-World Benchmarking. To verify the sparse matrix $\mathcal{O}(|E|)$ execution without encountering unbounded memory explosions, we benchmarked the metric on the SNAP Epinions dataset (`soc-sign-epinions`), containing over 131,000 nodes and 840,000 directed edges (projected symmetrically for evaluation). The implementation scaled smoothly, proving the $\mathcal{O}(d)$ degree normalization effectively contained the combinatorial cycle proliferation common in dense real-world topologies.